

Malkin Textbook Responses – Final Report

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Explain why the value θ_K , entering the Kohlrausch function, Eq. 2.1.6 is not a relaxation time. How do you find relaxation times for this relaxation function?

The Kohlrausch function is

$$\sigma(t) = \sigma_0 \exp \left[- \left(\frac{t}{\theta_K} \right)^n \right].$$

Therefore the relaxation function is $\varphi(t) = \exp \left[- \left(\frac{t}{\theta_K} \right)^n \right]$. It can also be expressed in terms of is the relaxation-time spectrum $G(\theta)$.

$$\varphi(t) = \int_0^\infty G(\theta) e^{-t/\theta} d\theta. \quad (1.1)$$

This gives us the weighting for each relaxation time which is what we need to solve this problem. To begin, we will do a change of variables. Let $s = 1/\theta$, so $d\theta = -ds/s^2$, then

$$\varphi(t) = \int_0^\infty \frac{G(1/s)}{s^2} e^{-st} ds \equiv \int_0^\infty \tilde{G}(s) e^{-st} ds, \quad \tilde{G}(s) = \frac{G(1/s)}{s^2}.$$

Therefore $\tilde{G}$ is obtained by an inverse Laplace transform:

$$\begin{aligned} \tilde{G}(s) &= \mathcal{L}^{-1}\{\varphi(t)\}(s) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} e^{st} \varphi(t) dt \\ \tilde{G}(s) &= \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} e^{st} e^{-(s_K t)^n} dt, \quad \text{where } s_K = 1/\theta_K \end{aligned}$$

1.1 Discrete relaxation times

If we want a simple solution, we can assume a discrete set of relaxation times. We can write

$$G(\theta) = \sum_{i=1}^N G_i \delta(\theta - \theta_i)$$

so that there are N discrete relaxation times θ_i with weights G_i . Then we have our relaxation function as:

$$\varphi(t) = \sum_{i=1}^N \int_0^\infty G_i \delta(\theta - \theta_i) e^{-t/\theta} d\theta = \sum_{i=1}^N G_i e^{-t/\theta_i}.$$

Now we can equate this to the Kohlrausch function:

$$e^{-(t/\theta_K)^n} = \sum_{i=1}^N G_i e^{-t/\theta_i} \implies \sigma(t) = \sigma_0 \sum_{i=1}^N G_i e^{-t/\theta_i}$$

1.2 Analytic solution

Following Pollard (1946)¹, the inverse Laplace transform of

$$\tilde{G}(s) = \frac{1}{2\pi i} \int_{\gamma} e^{st} e^{-(s_K t)^n} dt$$

can be simplified to

$$\tilde{G}(s) = \frac{1}{\pi} \operatorname{Im} \left\{ \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k!} e^{-i\pi n k} \frac{\Gamma(nk + 1)}{(s/s_K)^{nk+1}} \right\} \cdot \frac{1}{s_K}. \quad (1.2)$$

The relaxation spectrum is therefore $G(\theta) = \tilde{G}(1/\theta)/\theta^2$:

$$G(\theta) = \frac{1}{\pi \theta} \operatorname{Im} \left\{ \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k!} e^{-i\pi n k} \Gamma(nk + 1) \left(\frac{\theta}{\theta_K} \right)^{nk} \right\}.$$

Now we can verify this result by substituting $G(\theta)$ back into equation 1.1 and confirming that the integral recovers the Kohlrausch function. For case $n = 1/2$ (this is shown in figure 1) the numerical integration of equation 1.2 is able to reproduce $\varphi(t) = e^{-\sqrt{t/\theta_K}}$. For higher values of n I was not able to reproduce this due to numerical instabilities from the Gamma function.

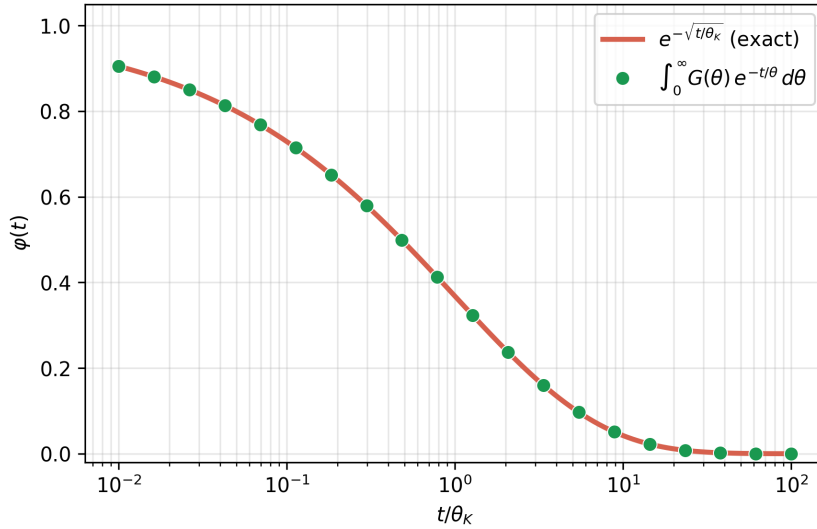


Figure 1: The relaxation function $\varphi(t) = e^{-\sqrt{t/\theta_K}}$ (solid line) replicated by the numerical integration of equation 1.2.

¹Pollard, H. (1946). The representation of e^{-x^λ} as a Laplace integral.

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Eq. 2.3.11 and its solution show that the Burgers model describes the behavior of a material with two relaxation times. The same behavior is represented by two parallel Maxwell elements with their relaxation times $\theta_1 = \eta_1/G_1$ and $\theta_2 = \eta_2/G_2$ where η_1 and G_1 are the viscosity and elastic modulus of the first and η_2 and G_2 of the second Maxwell elements joined in parallel. Calculate the values of the constants of the Burgers model expressed via constants of the two Maxwell elements.

2.1 Burgers Model:

For the total strain rate we have:

$$\dot{\gamma} = \dot{\gamma}_m + \dot{\gamma}_k = \frac{\sigma}{\eta_m} + \frac{\dot{\sigma}}{G_m} + \dot{\gamma}_k.$$

$$\dot{\gamma}_k = \dot{\gamma} - \frac{\sigma}{\eta_m} - \frac{\dot{\sigma}}{G_m}, \quad \ddot{\gamma}_k = \ddot{\gamma} - \frac{\dot{\sigma}}{\eta_m} - \frac{\ddot{\sigma}}{G_m}.$$

The stress across the Maxwell and Kelvin-Voigt elements is the same:

$$\sigma = \sigma_m = \sigma_k \implies \sigma = G_k \gamma_k + \eta_k \dot{\gamma}_k.$$

Now we can differentiate this then substitute Eq. 2.1 into it:

$$\dot{\sigma} = G_k \dot{\gamma}_k + \eta_k \ddot{\gamma}_k.$$

$$\dot{\sigma} = G_k \left(\dot{\gamma} - \frac{\sigma}{\eta_m} - \frac{\dot{\sigma}}{G_m} \right) + \eta_k \left(\ddot{\gamma} - \frac{\dot{\sigma}}{\eta_m} - \frac{\ddot{\sigma}}{G_m} \right).$$

Collecting terms gives:

$$\frac{\eta_k}{G_m} \ddot{\sigma} + \left(1 + \frac{G_k}{G_m} + \frac{\eta_k}{\eta_m} \right) \dot{\sigma} + \frac{G_k}{\eta_m} \sigma = \eta_k \ddot{\gamma} + G_k \dot{\gamma}. \quad (2.1)$$

2.2 Two Maxwell Elements in Parallel:

Consider branch 1, a Maxwell element with spring G_1 in series with dashpot η_1 . Within this Maxwell element the stress is the same in the spring and dashpot:

$$\sigma_1 = \sigma_{G_1} = \sigma_{\eta_1}, \quad \sigma_1 = G_1 \gamma_1, \quad \sigma_1 = \eta_1 \dot{\gamma}_1.$$

Therefore,

$$\dot{\gamma}_{G_1} = \frac{\dot{\sigma}_1}{G_1}, \quad \dot{\gamma}_{\eta_1} = \frac{\sigma_1}{\eta_1}.$$

Since the spring and dashpot are in series, strains add in branch 1:

$$\begin{aligned} \dot{\gamma}_1 &= \dot{\gamma}_{G_1} + \dot{\gamma}_{\eta_1} \\ &= \frac{\dot{\sigma}_1}{G_1} + \frac{\sigma_1}{\eta_1}. \end{aligned}$$

Let $D \equiv \frac{d}{dt}$, this may be written as

$$\dot{\gamma}_1 = \left(\frac{1}{G_1} D + \frac{1}{\eta_1} \right) \sigma_1 = \frac{\eta_1 D + G_1}{G_1 \eta_1} \sigma_1.$$

Similarly for branch 2:

$$\dot{\gamma}_2 = \frac{\dot{\sigma}_2}{G_2} + \frac{\sigma_2}{\eta_2} = \left(\frac{1}{G_2} D + \frac{1}{\eta_2} \right) \sigma_2 = \frac{\eta_2 D + G_2}{G_2 \eta_2} \sigma_2.$$

Since the two branches are in parallel, the total strain rate is the same in both branches:

$$\dot{\gamma} = \frac{\eta_1 D + G_1}{G_1 \eta_1} \sigma_1 = \frac{\eta_2 D + G_2}{G_2 \eta_2} \sigma_2.$$

$$\sigma_1 = \frac{G_1 \eta_1}{\eta_1 D + G_1} \dot{\gamma}, \quad \sigma_2 = \frac{G_2 \eta_2}{\eta_2 D + G_2} \dot{\gamma}.$$

The total stress is the sum of the stresses in each branch:

$$\begin{aligned} \sigma &= \sigma_1 + \sigma_2 \\ \sigma &= \frac{G_1 \eta_1}{\eta_1 D + G_1} \dot{\gamma} + \frac{G_2 \eta_2}{\eta_2 D + G_2} \dot{\gamma} \\ \sigma &= \left(\frac{G_1 \eta_1}{\eta_1 D + G_1} + \frac{G_2 \eta_2}{\eta_2 D + G_2} \right) \dot{\gamma} \\ \sigma &= \frac{G_1 \eta_1 (\eta_2 D + G_2) + G_2 \eta_2 (\eta_1 D + G_1)}{(\eta_2 D + G_2)(\eta_1 D + G_1)} \dot{\gamma} \\ (\eta_2 D + G_2)(\eta_1 D + G_1) \sigma &= (G_1 \eta_1 (\eta_2 D + G_2) + G_2 \eta_2 (\eta_1 D + G_1)) \dot{\gamma} \\ (\eta_1 \eta_2 D^2 + (\eta_1 G_2 + \eta_2 G_1) D + G_1 G_2) \sigma &= (G_1 \eta_1 \eta_2 D + G_1 \eta_1 G_2 + G_2 \eta_2 \eta_1 D + G_2 \eta_2 G_1) \dot{\gamma} \\ \eta_1 \eta_2 \ddot{\sigma} + (\eta_1 G_2 + \eta_2 G_1) \dot{\sigma} + G_1 G_2 \sigma &= ((G_1 \eta_1 \eta_2 + G_2 \eta_2 \eta_1) D + G_1 \eta_1 G_2 + G_2 \eta_2 G_1) \dot{\gamma} \\ \eta_1 \eta_2 \ddot{\sigma} + (\eta_1 G_2 + \eta_2 G_1) \dot{\sigma} + G_1 G_2 \sigma &= (G_1 \eta_1 \eta_2 + G_2 \eta_2 \eta_1) \ddot{\gamma} + (G_1 \eta_1 G_2 + G_2 \eta_2 G_1) \dot{\gamma} \end{aligned}$$

Finally,

$$\frac{\eta_1 \eta_2}{G_1 G_2} \ddot{\sigma} + \frac{(\eta_1 G_2 + \eta_2 G_1)}{G_1 G_2} \dot{\sigma} + \sigma = \frac{\eta_1 \eta_2 (G_1 + G_2)}{G_1 G_2} \ddot{\gamma} + (\eta_1 + \eta_2) \dot{\gamma} \quad (2.2)$$

2.3 Equating Coefficients

From Eq. 2.2 we have

$$\frac{\eta_1 \eta_2}{G_1 G_2} \ddot{\sigma} + \frac{(\eta_1 G_2 + \eta_2 G_1)}{G_1 G_2} \dot{\sigma} + \sigma = \frac{\eta_1 \eta_2 (G_1 + G_2)}{G_1 G_2} \ddot{\gamma} + (\eta_1 + \eta_2) \dot{\gamma}$$

From Eq. 2.1 we have

$$\frac{\eta_k}{G_m} \ddot{\sigma} + \left(1 + \frac{G_k}{G_m} + \frac{\eta_k}{\eta_m} \right) \dot{\sigma} + \frac{G_k}{\eta_m} \sigma = \eta_k \ddot{\gamma} + G_k \dot{\gamma}.$$

²Thank you to Grecia for showing me this method.

So we can compare the two equations, we'll multiply Eq. 2.1 by $\frac{\eta_m}{G_k}$:

$$\frac{\eta_k \eta_m}{G_m G_k} \ddot{\sigma} + \left(\frac{\eta_m}{G_k} + \frac{\eta_m}{G_m} + \frac{\eta_k}{G_k} \right) \dot{\sigma} + \sigma = \frac{\eta_k \eta_m}{G_k} \ddot{\gamma} + \eta_m \dot{\gamma}.$$

From left to right, we equate coefficients:

$$\frac{\eta_1 \eta_2}{G_1 G_2} = \frac{\eta_k \eta_m}{G_m G_k} \quad (2.3)$$

$$\frac{(\eta_1 G_2 + \eta_2 G_1)}{G_1 G_2} = \frac{\eta_m}{G_k} + \frac{\eta_m}{G_m} + \frac{\eta_k}{G_k} \quad (2.4)$$

$$\frac{\eta_1 \eta_2 (G_1 + G_2)}{G_1 G_2} = \frac{\eta_k \eta_m}{G_k} \quad (2.5)$$

$$(\eta_1 + \eta_2) = \eta_m \quad (2.6)$$

Firstly we have the trivial result from Eq. 2.6:

$$\boxed{\eta_m = \eta_1 + \eta_2}.$$

Theoretically, we also have

$$G_m = G_1 + G_2$$

but we can show this by plugging Eq. 2.3 into Eq. 2.5:

$$\begin{aligned} \frac{\eta_1 \eta_2}{G_1 G_2} (G_1 + G_2) &= \frac{\eta_k \eta_m}{G_k} \\ \frac{\eta_k \eta_m}{G_m G_k} (G_1 + G_2) &= \frac{\eta_k \eta_m}{G_k} \end{aligned}$$

$$\boxed{G_m = G_1 + G_2}.$$

Using $\eta_m = \eta_1 + \eta_2$, $G_m = G_1 + G_2$, and

$$\frac{\eta_k}{G_k} = \frac{\eta_1 \eta_2 (G_1 + G_2)}{G_1 G_2 (\eta_1 + \eta_2)},$$

we can substitute into Eq. 2.4 to solve for G_k :

$$\begin{aligned} \frac{\eta_m}{G_k} &= \frac{\eta_1 G_2 + \eta_2 G_1}{G_1 G_2} - \frac{\eta_1 + \eta_2}{G_1 + G_2} - \frac{\eta_1 \eta_2 (G_1 + G_2)}{G_1 G_2 (\eta_1 + \eta_2)}. \\ \frac{\eta_m}{G_k} &= \frac{(\eta_1 G_2 + \eta_2 G_1)(G_1 + G_2)(\eta_1 + \eta_2) - G_1 G_2 (\eta_1 + \eta_2)^2 - \eta_1 \eta_2 (G_1 + G_2)^2}{G_1 G_2 (G_1 + G_2) (\eta_1 + \eta_2)}. \end{aligned}$$

Now expand the first term in the numerator:

$$\begin{aligned} (\eta_1 G_2 + \eta_2 G_1)(\eta_1 + \eta_2) &= \eta_1^2 G_2 + \eta_1 \eta_2 G_2 + \eta_1 \eta_2 G_1 + \eta_2^2 G_1 \\ &= \eta_1^2 G_2 + \eta_2^2 G_1 + \eta_1 \eta_2 (G_1 + G_2). \end{aligned}$$

So

$$\begin{aligned} (\eta_1 G_2 + \eta_2 G_1)(G_1 + G_2)(\eta_1 + \eta_2) &= (G_1 + G_2)[\eta_1^2 G_2 + \eta_2^2 G_1 + \eta_1 \eta_2 (G_1 + G_2)] \\ &= (G_1 + G_2)(\eta_1^2 G_2 + \eta_2^2 G_1) + \eta_1 \eta_2 (G_1 + G_2)^2. \end{aligned}$$

Substitute this back into the numerator, the $\eta_1 \eta_2 (G_1 + G_2)^2$ terms cancel:

$$\begin{aligned} \text{numerator} &= \left[(G_1 + G_2)(\eta_1^2 G_2 + \eta_2^2 G_1) + \eta_1 \eta_2 (G_1 + G_2)^2 \right] \\ &\quad - G_1 G_2 (\eta_1 + \eta_2)^2 - \eta_1 \eta_2 (G_1 + G_2)^2 \\ &= (G_1 + G_2)(\eta_1^2 G_2 + \eta_2^2 G_1) - G_1 G_2 (\eta_1 + \eta_2)^2. \end{aligned}$$

Expand the remaining pieces:

$$\begin{aligned} (G_1 + G_2)(\eta_1^2 G_2 + \eta_2^2 G_1) &= \eta_1^2 G_1 G_2 + \eta_1^2 G_2^2 + \eta_2^2 G_1^2 + \eta_2^2 G_1 G_2, \\ G_1 G_2 (\eta_1 + \eta_2)^2 &= G_1 G_2 (\eta_1^2 + 2\eta_1 \eta_2 + \eta_2^2) = \eta_1^2 G_1 G_2 + 2\eta_1 \eta_2 G_1 G_2 + \eta_2^2 G_1 G_2. \end{aligned}$$

Subtracting gives

$$\text{numerator} = (\eta_1^2 G_2^2 + \eta_2^2 G_1^2) - 2\eta_1 \eta_2 G_1 G_2 = (G_1 \eta_2 - G_2 \eta_1)^2.$$

then

$$\frac{\eta_m}{G_k} = \frac{(G_1 \eta_2 - G_2 \eta_1)^2}{G_1 G_2 (G_1 + G_2) (\eta_1 + \eta_2)}.$$

Since $\eta_m = \eta_1 + \eta_2$, invert to obtain

$$G_k = \frac{G_1 G_2 (G_1 + G_2) (\eta_1 + \eta_2)^2}{(G_1 \eta_2 - G_2 \eta_1)^2}.$$

Finally, using Eq. 2.3 $\eta_k/G_k = \eta_1 \eta_2 (G_1 + G_2)/[G_1 G_2 (\eta_1 + \eta_2)]$, we can solve for η_k :

$$\eta_k = G_k \frac{\eta_k}{G_k} = \left[\frac{G_1 G_2 (G_1 + G_2) (\eta_1 + \eta_2)^2}{(G_1 \eta_2 - G_2 \eta_1)^2} \right] \left[\frac{\eta_1 \eta_2 (G_1 + G_2)}{G_1 G_2 (\eta_1 + \eta_2)} \right],$$

so

$$\eta_k = \frac{\eta_1 \eta_2 (G_1 + G_2)^2 (\eta_1 + \eta_2)}{(G_1 \eta_2 - G_2 \eta_1)^2}.$$

3 3-5 and 3-6 Malkin

3-5: Calculate shear stresses in the flow of liquid through a straight tube if the flow is created by the pressure gradient $\Delta p/L$ (L is the length of a tube).

3-6: Calculate the radial distribution of shear rates and flow velocity of Newtonian liquid (having viscosity η) through a straight tube with radius R .

I decided to combine these because the first part of 3-6 entirely covers 3-5.

Starting with the *Navier-Stokes equation*:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) = -\nabla p + \nabla \cdot \boldsymbol{\sigma}.$$

Evaluating the left-hand side, we have the following due to the steady flow assumption:

$$\frac{\partial \mathbf{u}}{\partial t} = 0.$$

And for the other LHS term:

$$((\mathbf{u} \cdot \nabla) \mathbf{u})_i = u_j \partial_j u_i = u_r \partial_r u_i + u_\theta \frac{1}{r} \partial_\theta u_i + u_z \partial_z u_i$$

$u_r = u_\theta = 0$ therefore

$$((\mathbf{u} \cdot \nabla) \mathbf{u})_r = ((\mathbf{u} \cdot \nabla) \mathbf{u})_\theta = 0$$

$\mathbf{u} = u_z(r)$

$$((\mathbf{u} \cdot \nabla) \mathbf{u})_i = u_r \partial_r u_z + u_\theta \frac{1}{r} \partial_\theta u_z + u_z \partial_z u_z = 0$$

Therefore the whole LHS of the Navier Stokes is zero. Navier-Stokes equation reduces to

$$\nabla p = \nabla \cdot \boldsymbol{\sigma}.$$

In cylindrical coordinates, the RHS is

$$\nabla \cdot \boldsymbol{\sigma} = \frac{1}{r} \frac{\partial}{\partial r} (r \sigma_{rz}) + \frac{1}{r} \frac{\partial \sigma_{\theta z}}{\partial \theta} + \frac{\partial \sigma_{zz}}{\partial z} = \frac{1}{r} \frac{\partial}{\partial r} (r \sigma_{rz}).$$

All other terms are zero since $\sigma_{\theta z} = 0$ and σ_{zz} is independent of z for steady flow. The pressure gradient is constant, so $\nabla p = -\frac{\Delta P}{L}$, and we have

$$\begin{aligned} -\frac{\Delta P}{L} &= \frac{1}{r} \frac{\partial}{\partial r} (r \sigma_{rz}) \\ \frac{\Delta P}{L} r &= -\frac{\partial}{\partial r} (r \sigma_{rz}) \\ \int \frac{\Delta P}{L} r dr &= -\int \frac{\partial}{\partial r} (r \sigma_{rz}) dr \\ \frac{\Delta P}{L} \frac{r^2}{2} + C_1 &= -r \sigma_{rz}. \end{aligned}$$

Solving for the shear stress gives

$$\sigma_{rz} = -\frac{\Delta P}{2L} r - \frac{C_1}{r}.$$

The stress must remain finite at the centerline $r = 0$, which requires $C_1 = 0$. Therefore

$$\boxed{\sigma_{rz} = -\frac{\Delta P}{2L} r}.$$

This is a key result: the shear stress σ_{rz} is proportional to the radius r and does not depend on the viscosity η .

Now we can use the constitutive relation for a Newtonian fluid $\sigma_{rz} = \eta \dot{\gamma}_{rz}$ to find the shear rate distribution:

$$\dot{\gamma}_{rz} = \frac{\sigma_{rz}}{\eta} = -\frac{\Delta P}{2\eta L} r.$$

Integrating the shear rate gives the velocity distribution:

$$\begin{aligned} u_z(r) &= \int \dot{\gamma}_{rz} dr \\ &= -\int \frac{\Delta P}{2\eta L} r dr \\ &= -\frac{\Delta P}{4\eta L} r^2 + C_2. \end{aligned}$$

Applying the no-slip boundary condition $u_z(R) = 0$ gives $C_2 = +\frac{\Delta P}{4\eta L} R^2$, so the velocity profile becomes

$$u_z(r) = \frac{\Delta P}{4\eta L} (R^2 - r^2).$$

Additional question 1. Calculate the volume output, Q , for the flow of Newtonian liquid.

The flow rate is given by

$$Q = \int \mathbf{u} \cdot d\mathbf{S}.$$

All of the flow field $\mathbf{u}$ has the same direction as the outlet S so

$$Q = \int u dS.$$

The surface is just the cross sectional area of the pipe:

$$Q = \int_0^R \int_0^{2\pi} u r dr d\theta = 2\pi \int_0^R u r dr$$

$$Q = 2\pi \frac{\Delta P}{4\eta L} \int_0^R (R^2 - r^2) r dr$$

$$R^2 \int_0^R r dr = \frac{R^4}{2}, \quad \int_0^R r^3 dr = \frac{R^4}{4}$$

$$Q = 2\pi \frac{\Delta P}{4\eta L} \left(\frac{R^4}{2} - \frac{R^4}{4} \right) = \boxed{\frac{\pi \Delta P R^4}{8\eta L}}$$

Additional question 2. Express maximum shear rate via volume output. From earlier we have:

$$|\sigma_{rz}| = \frac{\Delta P r}{2L}.$$

$r \in [0, R]$, therefore:

$$\sigma_{\max} = \frac{\Delta P R}{2L}.$$

$$Q = \frac{\pi R^3}{4\eta} \frac{\Delta P R}{2L} = \frac{\pi R^3}{4\eta} \sigma_{\max} \implies \sigma_{\max} = \frac{4\eta Q}{\pi R^3}$$

$$\dot{\gamma}_{\max} = \sigma_{\max}/\eta = \boxed{\frac{4Q}{\pi R^3}}$$

Additional question 3. *Is the last expression valid for a liquid with arbitrary rheological properties?*

No, in our derivation we assumed that $\sigma \propto \dot{\gamma}$. If we violate that assumption, we would need to modify our equation accordingly.

4 3-9 Malkin

Analyze the flow of a viscoplastic (“Bingham-type”) liquid through a straight tube of radius R . Find radial stress and velocity distributions and calculate volume output as a function of the pressure gradient.

According to Malkin, the *Bingham Equation* is

$$\sigma = \sigma_Y + \mu \dot{\gamma}, \quad (4.1)$$

where σ_Y is the yield stress and μ is the plastic viscosity. Or equivalently,

$$\dot{\gamma} = \begin{cases} 0 & \text{if } \sigma < \sigma_Y \\ \frac{\sigma - \sigma_Y}{\mu} & \text{if } \sigma \geq \sigma_Y \end{cases}.$$

The stress distribution follows the same argument as in 3-5/3-6, giving

$$\sigma_{rz} = \frac{\Delta P}{2L} r.$$

Flow only occurs when $\sigma(r) \geq \sigma_Y$, so there is a region in the center of the tube where $r < r_0$ for which there is no shearing. The radius r_0 is given by:

$$\sigma(r_0) = \sigma_Y \implies r_0 = \frac{2L}{\Delta P} \sigma_Y.$$

Recalling that for our “Bingham fluid” $\sigma = \sigma_Y + \mu \dot{\gamma}$, we can write $\dot{\gamma}$ as

$$\dot{\gamma} = \frac{\sigma - \sigma_Y}{\mu} = \frac{\Delta P}{2\mu L} r - \frac{\sigma_Y}{\mu}.$$

Adding the requirement that $\dot{\gamma} = 0$ for $r < r_0$ gives the full shear rate distribution:

$$\dot{\gamma} = \begin{cases} 0 & r < r_0 \\ \frac{\Delta P}{2\mu L}r - \frac{\sigma_Y}{\mu} & r \geq r_0 \end{cases}.$$

As you increase r , the velocity u_z decreases, so the shear rate $\dot{\gamma}$ is negative. Therefore,

$$\dot{\gamma} = -\frac{du_z}{dr} \implies du_z = -\dot{\gamma}dr.$$

Now we can integrate to get the velocity distribution in the region $r \geq r_0$:

$$u_z = -\int \dot{\gamma}dr = -\int \left(\frac{\Delta P}{2\mu L}r - \frac{\sigma_Y}{\mu} \right) dr = -\frac{\Delta P}{4\mu L}r^2 + \frac{\sigma_Y}{\mu}r + C.$$

From the zero slip condition at the wall, we have

$$0 = -\frac{\Delta P}{4\mu L}R^2 + \frac{\sigma_Y}{\mu}R + C \implies C = -\frac{\sigma_Y}{\mu}R + \frac{\Delta P}{4\mu L}R^2.$$

Therefore, the velocity distribution in the region $r \geq r_0$ is

$$u_z(r) = \frac{\Delta P}{4\mu L}(R^2 - r^2) - \frac{\sigma_Y}{\mu}(R - r).$$

For the region $r < r_0$, there is no shearing, so the velocity is constant. The value of this constant is given by the velocity at $r = r_0$:

$$u_z(r) = \frac{\Delta P}{4\mu L}(R^2 - r_0^2) - \frac{\sigma_Y}{\mu}(R - r_0).$$

Recall that

$$r_0 = \frac{2L}{\Delta P}\sigma_Y \implies \sigma_Y = \frac{\Delta P}{2L}r_0,$$

so

$$\frac{\sigma_Y}{\mu}(R - r_0) = \frac{\Delta P}{2\mu L}r_0(R - r_0).$$

Substituting this into the velocity expression gives

$$\begin{aligned} u_z(r) &= \frac{\Delta P}{4\mu L}(R^2 - r_0^2) - \frac{\Delta P}{2\mu L}r_0(R - r_0) = \frac{\Delta P}{4\mu L} [(R^2 - r_0^2) - 2r_0(R - r_0)] \\ &= \frac{\Delta P}{4\mu L}(R^2 - 2Rr_0 + r_0^2) = \frac{\Delta P}{4\mu L}(R - r_0)^2. \end{aligned}$$

Then the full velocity distribution is

$$u_z(r) = \begin{cases} \frac{\Delta P}{4\mu L}(R - r_0)^2 & r < r_0 \\ \frac{\Delta P}{4\mu L}(R^2 - r^2) - \frac{\sigma_Y}{\mu}(R - r) & r \geq r_0 \end{cases}.$$

Finally, we can calculate the volume flow Q by integrating the velocity profile across the cross-sectional area of the tube. Since $d\mathbf{A} = r dr d\theta \hat{z}$ and $\mathbf{u} = u_z(r)\hat{z}$, we have

$$\begin{aligned} Q &= 2\pi \int_0^R u_z(r) r dr \\ &= 2\pi \left[\int_0^{r_0} \frac{\Delta P}{4\mu L}(R - r_0)^2 r dr + \int_{r_0}^R \left(\frac{\Delta P}{4\mu L}(R^2 - r^2) - \frac{\sigma_Y}{\mu}(R - r) \right) r dr \right]. \end{aligned}$$

For the region $(0 \leq r \leq r_0)$:

$$\begin{aligned} \int_0^{r_0} \frac{\Delta P}{4\mu L}(R - r_0)^2 r dr &= \frac{\Delta P}{4\mu L}(R - r_0)^2 \int_0^{r_0} r dr \\ &= \frac{\Delta P}{4\mu L}(R - r_0)^2 \left[\frac{r^2}{2} \right]_0^{r_0} \\ &= \frac{\Delta P}{8\mu L}(R - r_0)^2 r_0^2. \end{aligned}$$

For the region $(r_0 \leq r \leq R)$:

$$\int_{r_0}^R \left(\frac{\Delta P}{4\mu L}(R^2 - r^2) - \frac{\sigma_Y}{\mu}(R - r) \right) r dr = \int_{r_0}^R \left[\frac{\Delta P}{4\mu L}(R^2 r - r^3) - \frac{\sigma_Y}{\mu}(Rr - r^2) \right] dr.$$

Integrate term-by-term:

$$\begin{aligned} \int R^2 r dr &= R^2 \frac{r^2}{2}, & \int r^3 dr &= \frac{r^4}{4}, \\ \int Rr dr &= R \frac{r^2}{2}, & \int r^2 dr &= \frac{r^3}{3}. \end{aligned}$$

Plugging the integrals back in gives

$$\begin{aligned} &\int_{r_0}^R \left(\frac{\Delta P}{4\mu L}(R^2 - r^2) - \frac{\sigma_Y}{\mu}(R - r) \right) r dr \\ &= \left[\frac{\Delta P}{4\mu L} \left(R^2 \frac{r^2}{2} - \frac{r^4}{4} \right) - \frac{\sigma_Y}{\mu} \left(R \frac{r^2}{2} - \frac{r^3}{3} \right) \right]_{r_0}^R. \end{aligned}$$

Evaluate at $r = R$:

$$\begin{aligned} &\frac{\Delta P}{4\mu L} \left(\frac{R^4}{2} - \frac{R^4}{4} \right) - \frac{\sigma_Y}{\mu} \left(\frac{R^3}{2} - \frac{R^3}{3} \right) \\ &= \frac{\Delta P}{16\mu L} R^4 - \frac{\sigma_Y}{6\mu} R^3. \end{aligned}$$

Evaluate at $r = r_0$:

$$\frac{\Delta P}{4\mu L} \left(R^2 \frac{r_0^2}{2} - \frac{r_0^4}{4} \right) - \frac{\sigma_Y}{\mu} \left(R \frac{r_0^2}{2} - \frac{r_0^3}{3} \right).$$

Combining both regions gives the final expression for Q :

$$Q = 2\pi \left[\frac{\Delta P}{8\mu L} (R - r_0)^2 r_0^2 + \frac{\Delta P}{16\mu L} R^4 - \frac{\sigma_Y}{6\mu} R^3 - \frac{\Delta P}{4\mu L} \left(R^2 \frac{r_0^2}{2} - \frac{r_0^4}{4} \right) + \frac{\sigma_Y}{\mu} \left(R \frac{r_0^2}{2} - \frac{r_0^3}{3} \right) \right]$$

Substituting $\sigma_Y = \frac{\Delta P}{2L} r_0$:

$$\begin{aligned} Q &= 2\pi \frac{\Delta P}{\mu L} \left[\frac{1}{8} (R - r_0)^2 r_0^2 + \frac{1}{16} R^4 - \frac{1}{12} R^3 r_0 - \frac{1}{4} \left(R^2 \frac{r_0^2}{2} - \frac{r_0^4}{4} \right) + \frac{1}{2} r_0 \left(R \frac{r_0^2}{2} - \frac{r_0^3}{3} \right) \right] \\ &= 2\pi \frac{\Delta P}{\mu L} \left[\frac{1}{8} R^2 r_0^2 - \frac{1}{4} R r_0^3 + \frac{1}{8} r_0^4 + \frac{1}{16} R^4 - \frac{1}{12} R^3 r_0 - \frac{1}{8} R^2 r_0^2 + \frac{1}{16} r_0^4 + \frac{1}{4} R r_0^3 - \frac{1}{6} r_0^4 \right] \\ &= 2\pi \frac{\Delta P}{\mu L} \left[\cancel{\frac{1}{8} R^2 r_0^2} - \cancel{\frac{1}{8} R^2 r_0^2} + \cancel{\frac{1}{4} R r_0^3} - \cancel{\frac{1}{4} R r_0^3} + \frac{1}{8} r_0^4 + \frac{1}{16} R^4 - \frac{1}{12} R^3 r_0 + \frac{1}{16} r_0^4 - \frac{1}{6} r_0^4 \right] \\ &= \frac{\pi \Delta P}{\mu L} \left[\frac{1}{8} R^4 - \frac{1}{6} R^3 r_0 + \frac{1}{24} r_0^4 \right] \end{aligned}$$

gives our final equation for Q in terms of the pressure gradient, the viscosity, the length and the critical radius (r_0).

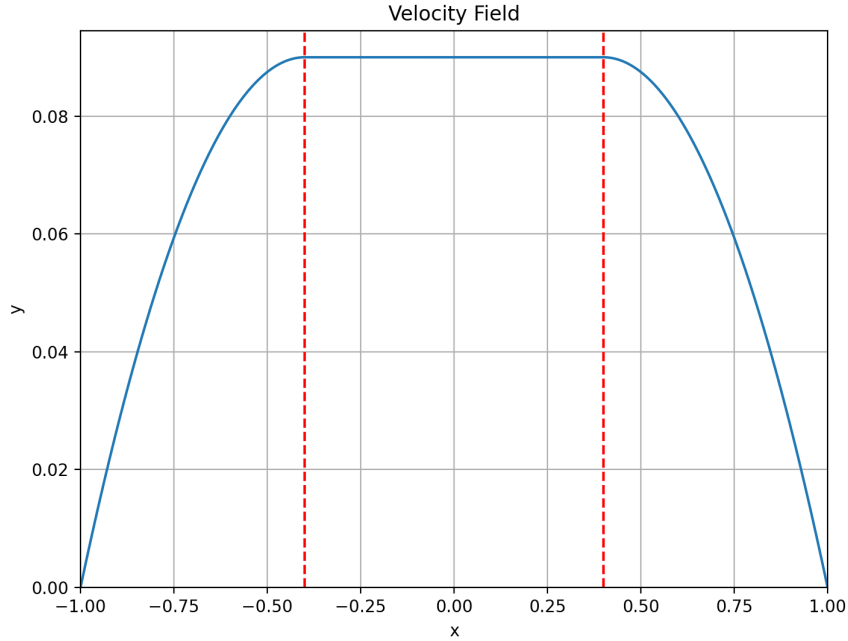


Figure 2: Velocity field for a Bingham fluid flowing through a tube.

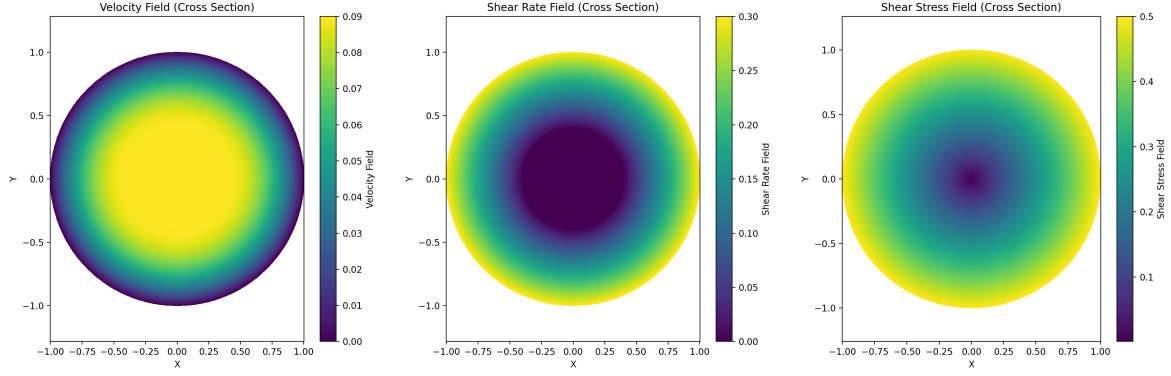


Figure 3: Cross-sectional profiles of velocity, shear rate, and shear stress for a Bingham fluid flowing through a tube. The region near the center of the tube (where $r < r_0$) has zero shear rate. The velocity profile is flat in this region and then decreases towards the walls of the tube. The shear stress increases linearly from $r = 0$ to $r = R$.

5 5-7 Malkin

In the principal scheme of measuring viscoelastic properties of the material, a spring is used (Fig. 5.7.1) that is not ideal elastic but viscoelastic (has some losses in deformation). How do you calculate the viscoelastic properties of the material under investigation?

The top plate is a damped harmonic oscillator being driven at an angular velocity of ω . The elastic and viscous components of the spring are captured by $Z^* = Z' + iZ''$ and the elastic and viscous components of the material between the plates are captured by $G^* = G' + iG''$. Its equation of motion is

$$m\ddot{x}_2 + Z^*x_2 + \frac{S}{h}G^*(x_2 - x_1) = 0 \quad (5.1)$$

where m is the mass of the top plate, S is the surface area of the plate, h is the height of the sample, x_2 is the displacement of the top plate and x_1 is the displacement of the driving plate. Since the system is linear the solution for x_1 and x_2 should have the same angular velocity ω :

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 e^{i\varphi} \end{bmatrix} e^{i\omega t}$$

Substituting $x_1 = c_1 e^{i\omega t}$, $x_2 = c_2 e^{i\varphi} e^{i\omega t}$ and $\ddot{x}_2 = -c_2 \omega^2 e^{i\varphi} e^{i\omega t}$ into equation 5.1:

$$-mc_2 \omega^2 e^{i\varphi} + c_2 Z^* e^{i\varphi} + \frac{S}{h} G^* (c_2 e^{i\varphi} - c_1) = 0$$

Solving for G^* :

$$\frac{S}{h} G^* (c_2 e^{i\varphi} - c_1) = mc_2 \omega^2 e^{i\varphi} - c_2 Z^* e^{i\varphi}$$

$$G^* = \frac{h}{S} \cdot \frac{m\omega^2 - Z^*}{1 - \frac{c_1}{c_2} e^{-i\varphi}}$$

For simplicity, let $r = c_1/c_2$. Separating real and imaginary components:

$$G^* = \frac{h}{S} \cdot \frac{m\omega^2 - Z' - iZ''}{(1 - r \cos \varphi) + ir \sin \varphi}$$

$$G^* = \frac{h}{S} \cdot \frac{[m\omega^2 - Z' - iZ''][(1 - r \cos \varphi) - ir \sin \varphi]}{[(1 - r \cos \varphi) + ir \sin \varphi][(1 - r \cos \varphi) - ir \sin \varphi]}$$

Just the numerator:

$$\begin{aligned} & [m\omega^2 - Z' - iZ''][(1 - r \cos \varphi) - ir \sin \varphi] \\ &= [m\omega^2 - Z' - iZ''](1 - r \cos \varphi) - [m\omega^2 - Z' - iZ'']ir \sin \varphi \\ &= (m\omega^2 - Z')(1 - r \cos \varphi) - iZ''(1 - r \cos \varphi) - [i(m\omega^2 - Z')r \sin \varphi + Z''r \sin \varphi] \\ &= (m\omega^2 - Z')(1 - r \cos \varphi) - iZ''(1 - r \cos \varphi) - i(m\omega^2 - Z')r \sin \varphi - Z''r \sin \varphi \\ &= (m\omega^2 - Z')(1 - r \cos \varphi) - Z''r \sin \varphi - i[(m\omega^2 - Z')r \sin \varphi + Z''(1 - r \cos \varphi)] \end{aligned}$$

Substituting the numerator back into G^* gives our final results:

$$G^* = \frac{h}{S} \cdot \frac{(m\omega^2 - Z')(1 - r \cos \varphi) - Z''r \sin \varphi - i[(m\omega^2 - Z')r \sin \varphi + Z''(1 - r \cos \varphi)]}{(1 - r \cos \varphi)^2 + r^2 \sin^2 \varphi}$$

$$G' = \frac{h}{S} \cdot \frac{(m\omega^2 - Z')(1 - r \cos \varphi) - Z''r \sin \varphi}{(1 - r \cos \varphi)^2 + r^2 \sin^2 \varphi}$$

$$G'' = -\frac{h}{S} \cdot \frac{(m\omega^2 - Z')r \sin \varphi + Z''(1 - r \cos \varphi)}{(1 - r \cos \varphi)^2 + r^2 \sin^2 \varphi}$$

$$\tan \delta = \frac{G''}{G'} = \frac{-[(m\omega^2 - Z')r \sin \varphi + Z''(1 - r \cos \varphi)]}{(m\omega^2 - Z')(1 - r \cos \varphi) - Z''r \sin \varphi}$$

According to Malkin, the results are:

$$G^* = \frac{(Z^* - m\omega^2)Be^{i\varphi}}{K(A - Be^{i\varphi})},$$

$$G' = \frac{(Z'' - m\omega^2)(p \cos \varphi - 1) - Z''p \sin \varphi}{K[(p - \cos \varphi)^2 + \sin^2 \varphi]},$$

$$G'' = \frac{(Z' - m\omega^2)p \sin \varphi + Z'(p \cos \varphi - 1)}{K[(p - \cos \varphi)^2 + \sin^2 \varphi]}.$$

These do not agree with my results. Putting aside differences in notation ($K = S/h$ and $A, B = c_1, c_2$), Malkin says that G' is a function of Z'' and not Z' and vice versa. Based on my calculations, I think this has to be a typo because if you swap one Z' with Z'' and another Z'' with Z' then our results agree. The denominators also look different but are really the same³.

³Malkin's denominator: $(p - \cos \varphi)^2 + \sin^2 \varphi = p^2 - 2p \cos \varphi + \cos^2 \varphi + \sin^2 \varphi = 1 + p^2 - 2p \cos \varphi$. Ours: $(1 - r \cos \varphi)^2 + r^2 \sin^2 \varphi = 1 - 2r \cos \varphi + r^2 \cos^2 \varphi + r^2 \sin^2 \varphi = 1 - 2r \cos \varphi + r^2$. These are equal since $r = p$.